

ASSIGNMENT: HOTEL MANAGEMENT

LECTURER: NHUNG NTH69

SUBJECT CODE: DBI202

MEMBERS

LE TRONG TAN

PHAN DANG GIA KHANG

TRAN QUANG TRONG

HUYNH PHUC KHANG

TRAN THANH TUNG

REASONS FOR CHOOSING THE TOPIC

RESEASON: THE TOURISM AND HOSPITALITY INDUSTRY IS CONSTANTLY EVOLVING. TRADITIONAL HOTEL MANAGEMENT METHODS (MANUAL RECORD-KEEPING, FRAGMENTED EXCEL SPREADSHEETS) ARE PRONE TO ERRORS SUCH AS OVERBOOKING, REVENUE LOSS, OR INCORRECT GUEST SERVICE ARRANGEMENTS.

PURPOSE: BUILDING THIS SYSTEM AUTOMATES THE PROCESS FROM GUEST BOOKING TO CHECK-OUT, ENABLING CENTRALIZED, ACCURATE, AND RAPID DATA MANAGEMENT.

INTRODUCTION

A HOTEL MANAGEMENT SYSTEM IS A DATABASE SYSTEM USED TO MANAGE HOTEL INFORMATION.

THE SYSTEM HELPS TO MANAGE:

- **CUSTOMERS**
- **ROOMS**
- **BOOKINGS**
- **SERVICES**
- **PAYMENTS**

THE GOAL IS TO STORE AND ORGANIZE HOTEL DATA EFFICIENTLY.

DATABASE STRUCTURE

THE SYSTEM CONTAINS 5 MAIN TABLES:

- **CUSTOMER**
- **ROOM**
- **BOOKINGS**
- **PAYMENT**
- **SERVICE**

THESE TABLES ARE CONNECTED USING PRIMARY KEYS AND FOREIGN KEYS.

CUSTOMER TABLE

THE CUSTOMER TABLE STORES CUSTOMER INFORMATION.

COLUMNS:

- **CUSTOMERID (PRIMARY KEY)**
- **NAME**
- **PHONE**
- **EMAIL**
- **IDCARD**

PURPOSE:

TO STORE CUSTOMER DETAILS WHEN THEY CHECK IN.

BOOKING TABLE

THE BOOKINGS TABLE MANAGES ROOM RESERVATIONS.

COLUMNS:

- **BOOKINGID (PRIMARY KEY)**
- **CUSTOMERID (FOREIGN KEY)**
- **ROOMID (FOREIGN KEY)**
- **CHECKINDATE**
- **CHECKOUTDATE**

PURPOSE:

TO RECORD WHICH CUSTOMER BOOKS WHICH ROOM.

PAYMENT TABLE

THE PAYMENT TABLE STORES PAYMENT INFORMATION.

COLUMNS:

- **PAYMENTID**
- **BOOKINGID**
- **AMOUNT**
- **PAYMENTDATE**

PURPOSE:

TO TRACK CUSTOMER PAYMENTS

SERVICE TABLE

THE SERVICE TABLE STORES HOTEL SERVICES.

COLUMNS:

- **SERVICEID**
- **SERVICENAME**
- **PRICE**

EXAMPLES:

- **LAUNDRY SERVICE**
- **BREAKFAST**
- **ROOM CLEANING**

STAFF TABLE

THE STAFF TABLE STORES INFORMATION ABOUT HOTEL EMPLOYEES.

COLUMNS:

- **STAFFID (PRIMARY KEY)**
- **FULLNAME**
- **ROLE**
- **PHONE**
- **USERNAME**
- **PASSWORDHASH**

PURPOSE:

THIS TABLE HELPS THE HOTEL MANAGE STAFF ACCOUNTS AND RESPONSIBILITIES.

EACH BOOKING CAN BE HANDLED BY A STAFF MEMBER.

INVOICE TABLE

THE INVOICES TABLE STORES BILLING INFORMATION FOR EACH BOOKING.

COLUMNS:

- **INVOICEID (PRIMARY KEY)**
- **BOOKINGID (FOREIGN KEY)**
- **TOTALAMOUNT**
- **CREATEDDATE**

PURPOSE:

THIS TABLE GENERATES THE FINAL BILL FOR THE CUSTOMER AFTER CHECKOUT.

PAYMENTS TABLE

THE PAYMENTS TABLE RECORDS PAYMENT TRANSACTIONS.

COLUMNS:

- **PAYMENTID**
- **INVOICEID**
- **AMOUNT**
- **PAYMENTDATE**
- **METHOD**

PURPOSE:

IT TRACKS HOW AND WHEN THE CUSTOMER PAID THE INVOICE.

BOOKINGDETAILS TABLE

THE BOOKINGDETAILS TABLE CONTAINS DETAILED ROOM BOOKING INFORMATION.

COLUMNS:

- BOOKINGDETAILID
- BOOKINGID
- ROOMID
- CHECKINDATE
- CHECKOUTDATE
- PRICEPERNIGHT

PURPOSE:

THIS TABLE STORES SPECIFIC ROOM BOOKING INFORMATION AND ALLOWS MULTIPLE ROOMS PER BOOKING.

ROOMPRICEHISTORY TABLE

THE ROOMPRICEHISTORY TABLE STORES HISTORICAL ROOM PRICES.

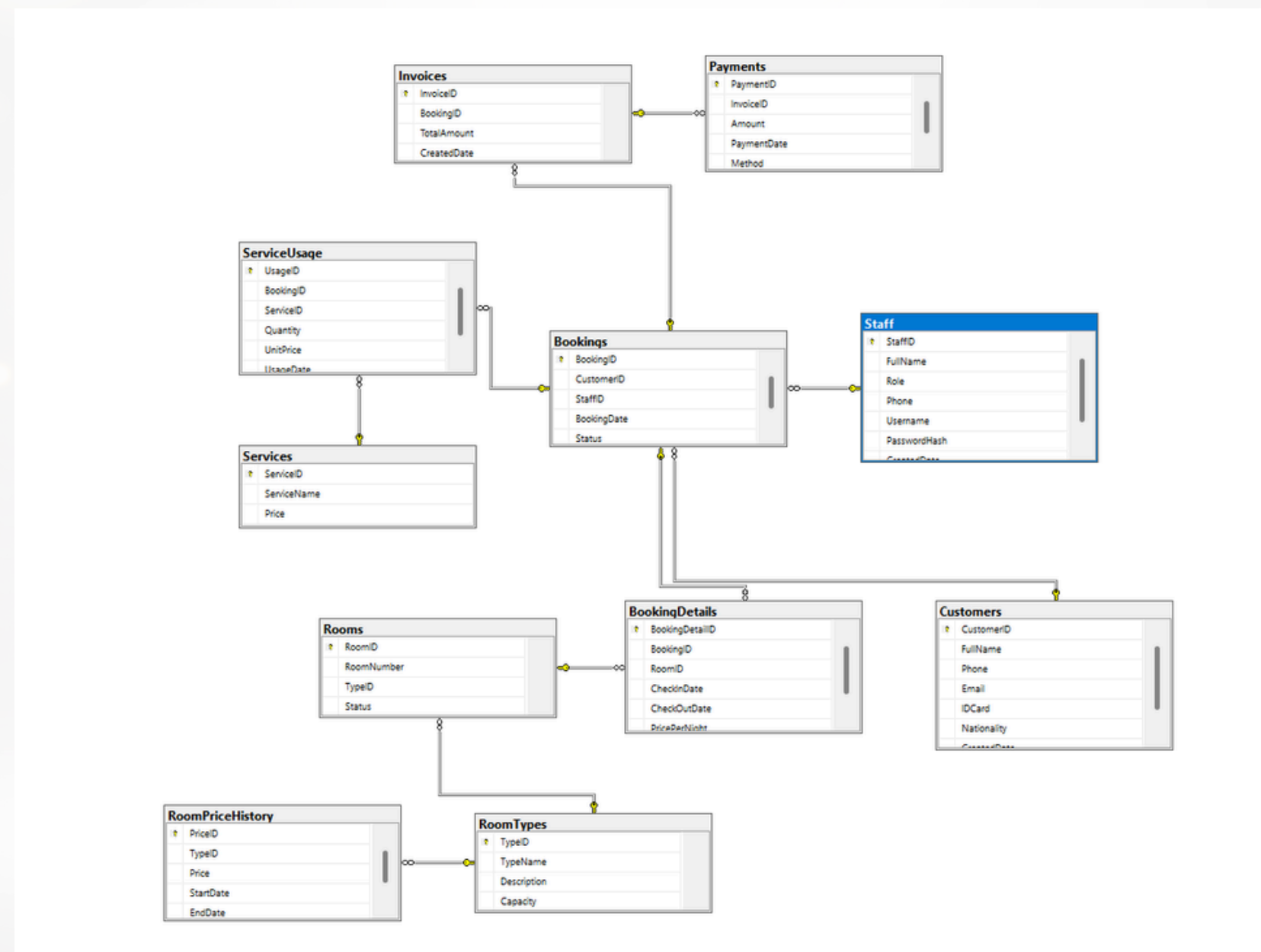
COLUMNS:

- PRICEID
- TYPEID
- PRICE
- STARTDATE
- ENDDATE

PURPOSE:

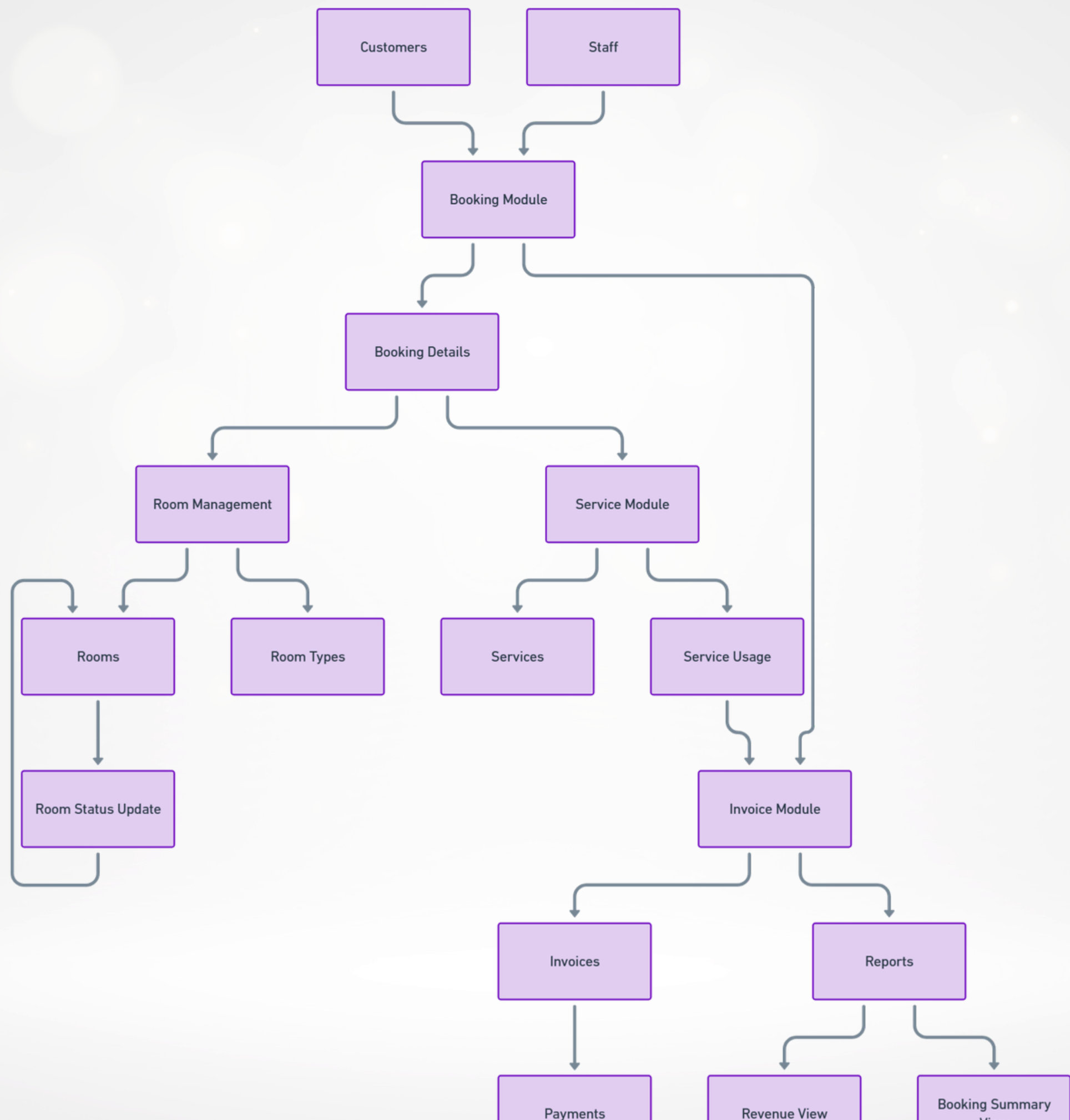
IT ALLOWS THE HOTEL TO TRACK ROOM PRICE CHANGES OVER TIME.

ENTITY RELATIONSHIP DIAGRAM



ADVANTAGES

- **EASY HOTEL DATA MANAGEMENT**
- **QUICK ROOM AVAILABILITY CHECK**
- **TRACK CUSTOMERS AND BOOKINGS**
- **AUTOMATIC ROOM STATUS UPDATE**



A top-down view of a clean, white desk. In the upper left, a white laptop is partially visible. To its right is a white spiral-bound notebook. A silver pen lies horizontally across the center. Below the pen, a pair of thin-framed glasses is visible. In the bottom left corner, a large, green, tropical-style leaf is partially shown. The overall lighting is bright and even.

**THANK YOU FOR
YOUR LISTENING**